PERTRONIC FIREMAP ETHERNET GATEWAY

USER MANUAL

AUSTRALIA & NEW ZEALAND

Issue 3.1

Valid for

PCB Hardware:	**v1.10, v1.15**
PCB Firmware:	**v3.02, v4.00.02**
FireMap®:	**v2.32, V3, V4, V5, V6, V7 or greater**

Note: The FireMap Ethernet Gateway (FMEGATE) was previously called the Ethernet Gateway Mk2 (NET2GATE). It was renamed to avoid confusion with Net2 Network products. The FireMap Gateway is NOT a Net2 Network product.

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1. INTRODUCTION

1.1 Function Description

The Ethernet Gateway (FMEGATE) enables Pertronic F100A and F120A Fire Panels to be remotely monitored and managed using Pertronic FireMap® Graphics System via LAN or Internet connections.

The FMEGATE is a 2-way interface between the Fire Panel and FireMap®. It collates and formats Fire Panel status information for use by FireMap® and forwards control commands (Zone / Device isolate, Alarm acknowledge etc) to the Fire Panel.

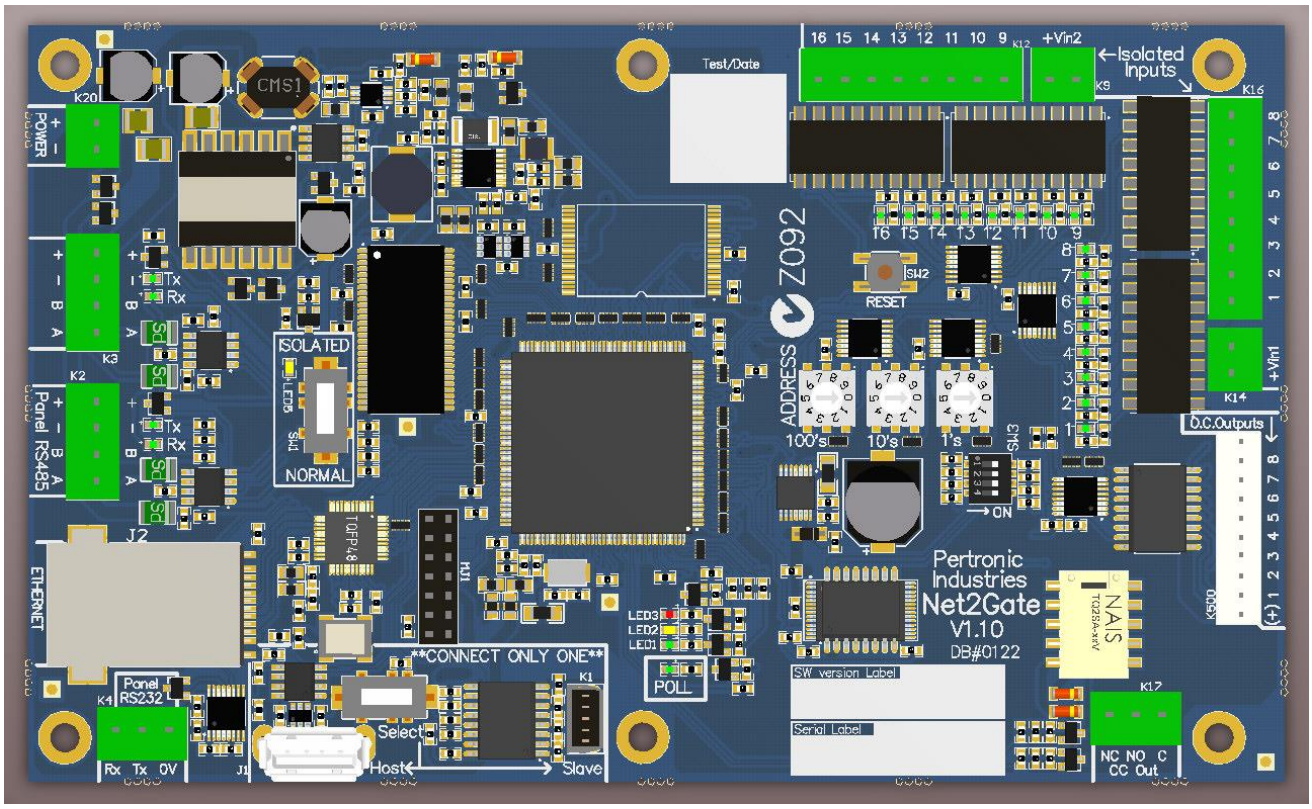
1.2 Features

The FMEGATE has the following features:

- Standard TCP/IP Ethernet Interface
- Panel RS485 to Ethernet conversion
- Web based configuration interface
- Low quiescent current
- IP, Netmask, Gateway address configurable via web interface
- 16 Digital inputs (interface for conventional Fire Panels or auxiliary equipment)
- Firmware upgrade via USB memory stick

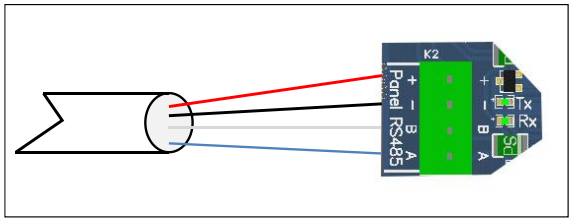
1.3 Specification

Electrical	**Operating Voltage** 18 – 30V	**Quiescent current** 50mA (at 27.4V) from 4-wire RS485 *plus* 6mA (at 27.4V) from external supply for each digital input.
Environmental	**Operating Temperature Range** -5ºC to 45ºC	**Humidity** 10 to 95% RH (non-condensing)≤40ºC ≤ 75% RH 41ºC to 45ºC
Mechanical	**Dimensions (H x W x D mm)** 27 x 165 x 100	**Weight** 125 g

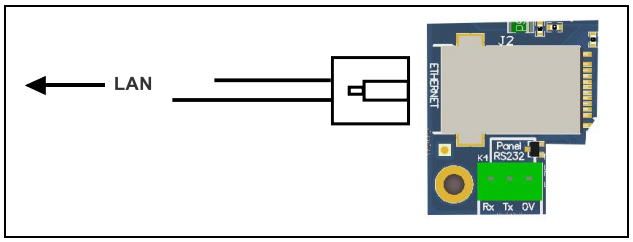

Test/Date
+Vin2
Isolated Inputs
K9
K16
K20
POWER
CMS1
Z092
ADDRESS
100's
10's
1's
SW3
ON
SW2
RESET
Tx
Rx
K3
K2
Panel RS485
ISOLATED
LED5
SW1
NORMAL
TQFP48
M1
J2
ETHERNET
K14
+Vin1
D.C.Outputs
K500
Pertronic Industries
Net2Gate
V1.10
DB#0122
NAIS
TQ2SA-xxV
LED3
LED2
LED1
POLL
SW version Label
Serial Label
K12
NC NO C
CC Out
Panel RS232
K4
Rx Tx 0V
J1
CONNECT ONLY ONE
K1
Select
Host
Slave

2. CONNECTIONS

2.1 RS485 Connections

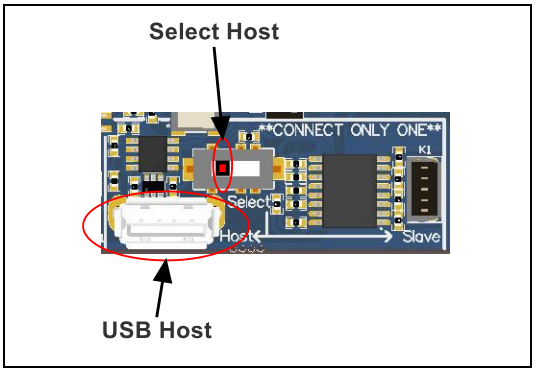


2.2 Ethernet Connection

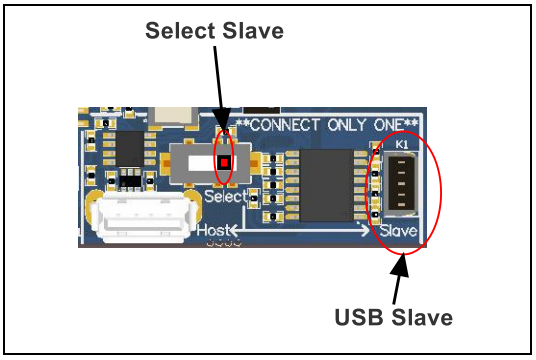


2.3 USB Connections

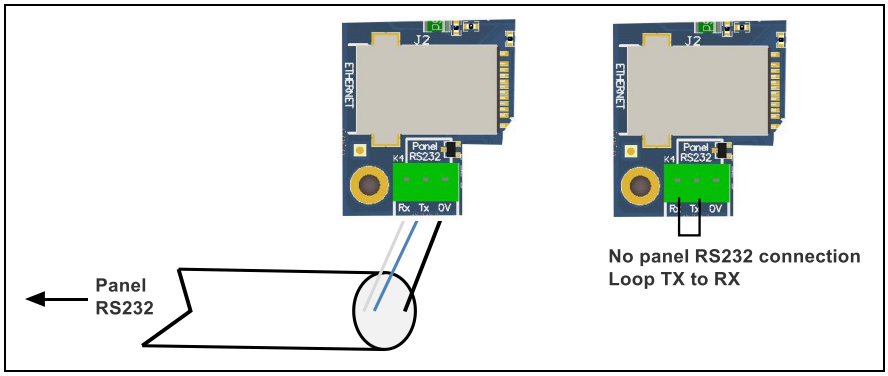
2.3.1 Host



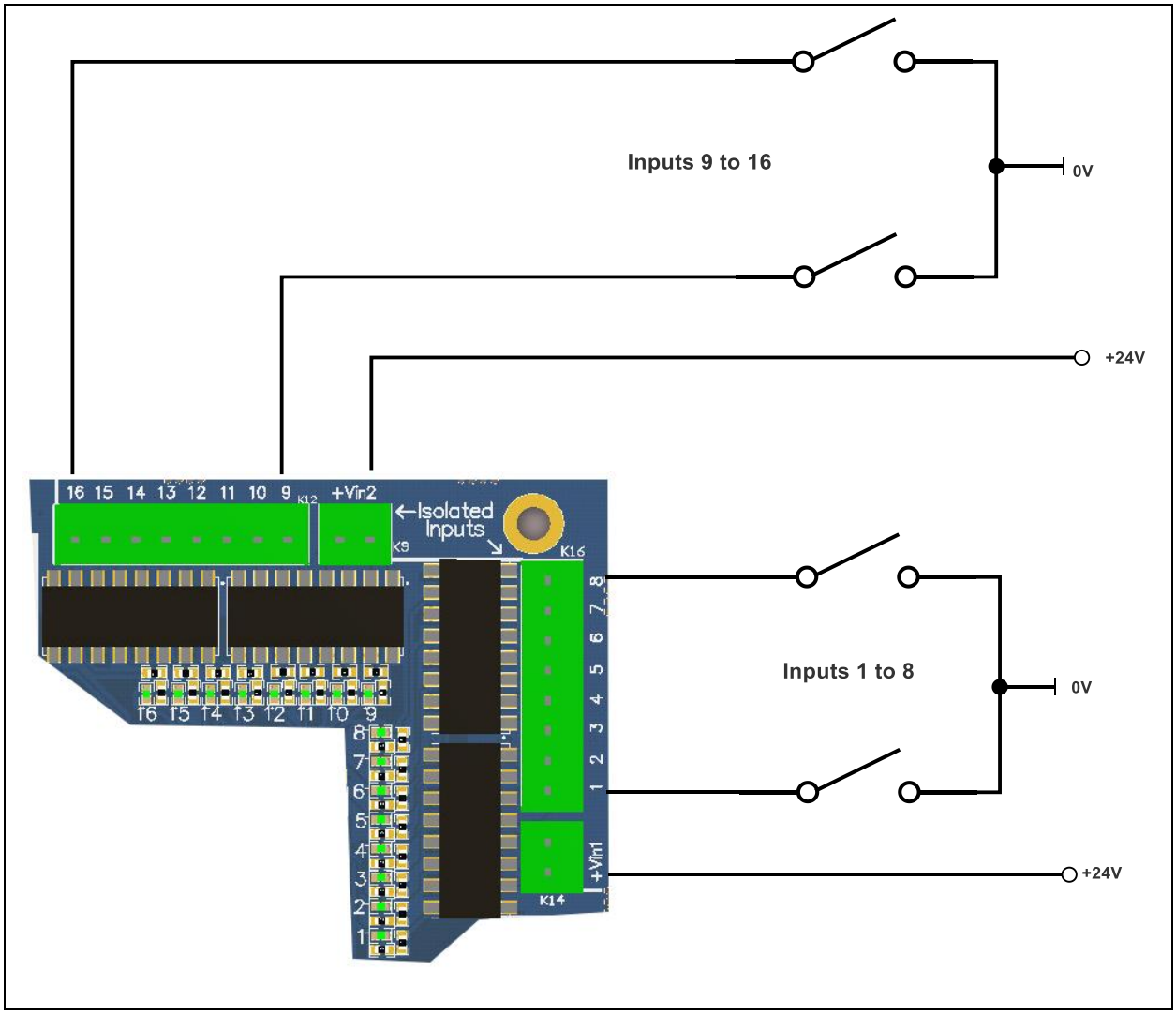
2.3.2 Slave (not used)



2.4 RS232



2.5 Isolated Digital Inputs



3. INSTALLATION

3.1 Operating Modes

It is possible to use some or all of the features offered by the FMEGATE. From the outset the installer should determine those features required and ensure that all connections are made during installation for the options selected. See the table below to determine the connections required.

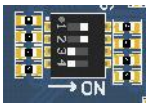
Options Required	Connections Required				
	Power	**RS485**	**RS232**	**Ethernet**	**Digital Inputs**
F100/120 basic		✓	**Note 1**	✓	
F100/120 + Digital Inputs		✓	**Note 1**	✓	✓
F100/120 + Detector PW		✓	✓	✓	
F100/120 + Detector PW + Digital Inputs		✓	✓	✓	✓
Digital Inputs Only (conventional panel etc.)	✓		**Note 1**	✓	✓

Note 1: Connect TX to RX if no RS232 connection is made to the Fire Panel

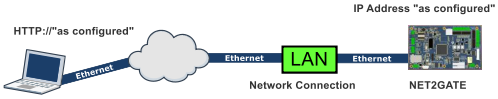
3.2 Initial Installation

The FMEGATE is designed to be installed within a Fire Panel cabinet. If it is not being used with a Fire Panel it should be housed in an equivalent auxiliary equipment cabinet.

Proceed with the following steps to complete installation:



1. Check that the FMEGATE dip-switch settings are set to factory default.
2. Mount the FMEGATE in the Fire Panel cabinet
3. Connect external wiring to Isolated Digital Inputs – see Section 2.5 (optional)
4. Connect RS485 or Power – see Section 2.1
5. Set up Laptop / PC for use with FMEGATE – see Section 3.3
6. Set FMEGATE IP address and configuration options – see Section 3.3.1 (optional)
7. Connect FMEGATE Ethernet port to its configured local network – see Section 2.2
8. Connect to the Configuration Interface in FMEGATE using a PC with access to the same local network FMEGATE is connected to. The IP address used must be the “configured” address (confirms FMEGATE can be accessed as configured).



9. Set up any remaining configuration options required before closing the connection from your computer.

3.3 Setting up your Laptop or PC

The configuration interface for FMEGATE is essentially a small website that the user accesses using their computer and web browser.

Until the FMEGATE has been configured with its “network settings” connection to the interface requires a physical cable connection between the user’s computer and FMEGATE Ethernet ports. After initial configuration and the FMEGATE is connected to its final network the configuration interface can be accessed by any computer that can connect to that network–simply by entering the configured IP address into the web browser.

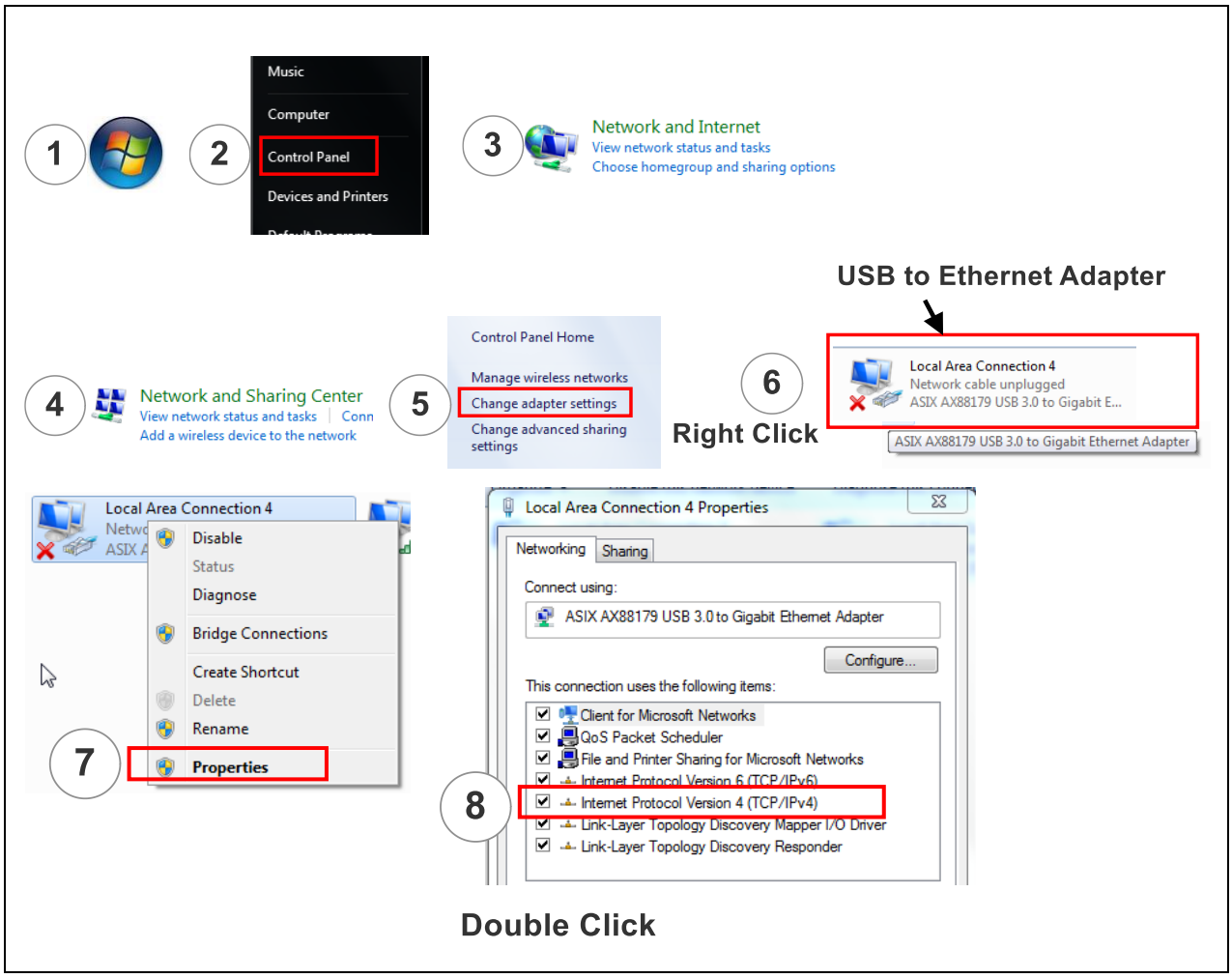
Unless your computer has been used previously to access a FMEGATE unit it will need to have configuration settings changed on the Ethernet port.

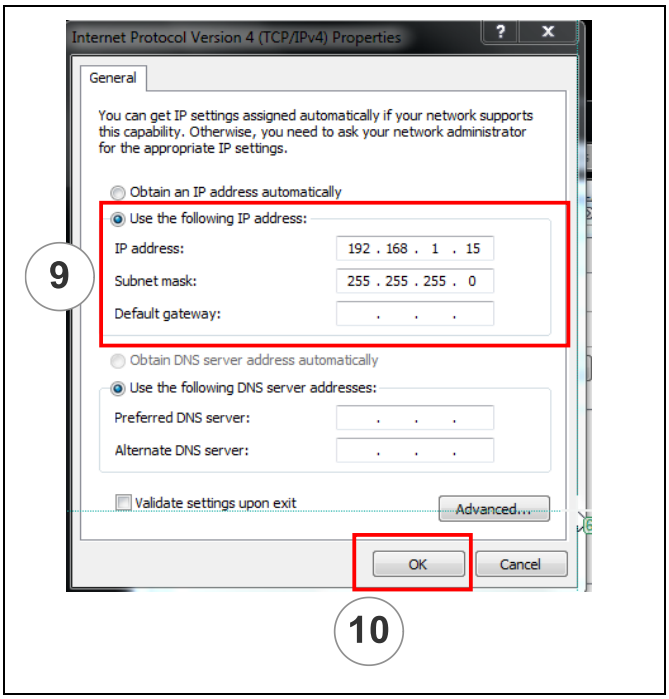
It is recommended that a second Ethernet port is set up using a USB to Ethernet adapter for the sole purpose of accessing devices requiring direct Ethernet cable connection–this removes the requirement to be constantly changing the primary Ethernet port settings for other uses.

3.3.1 Configuring the second Ethernet port (Recommended Option)

A suitable USB to Ethernet adapter is simply installed by plugging the device into a USB port and allowing the operating system to install a driver; either from a disk supplied with the unit or more commonly directly downloading the latest driver from the supplier's website.

Once the second Ethernet port is available it can be configured by following the steps illustrated below:

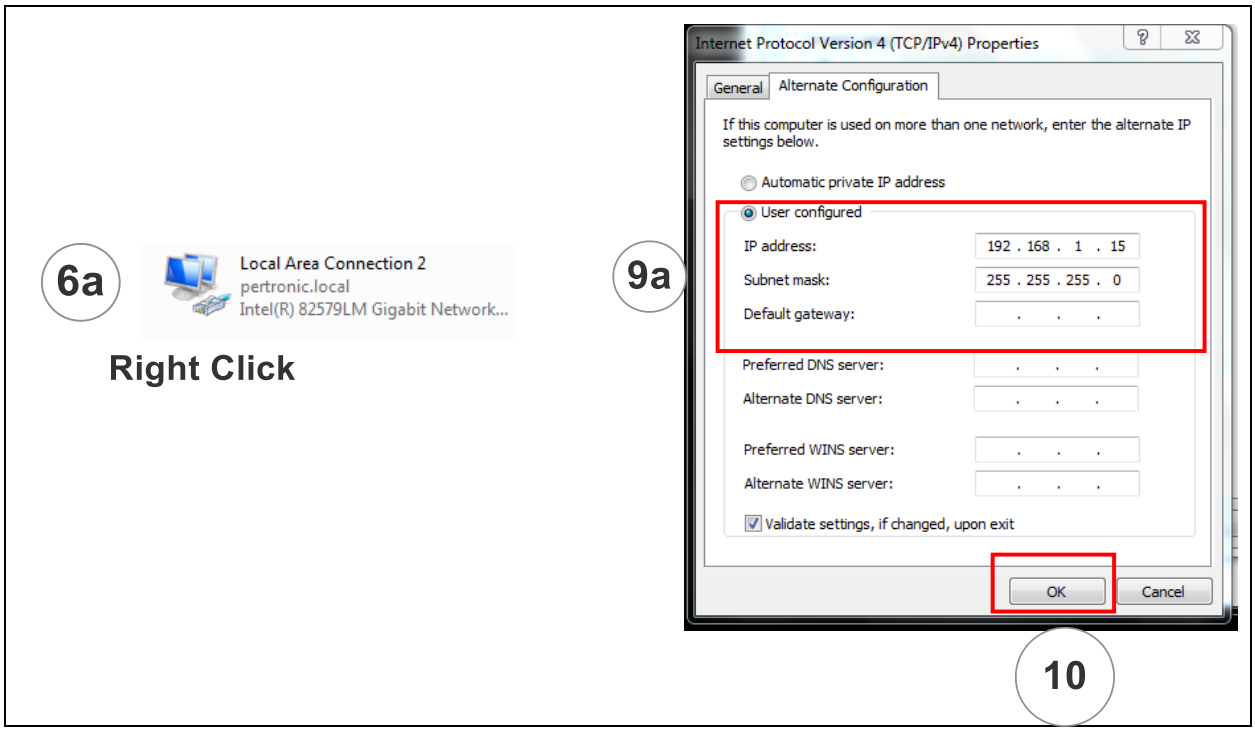




NOTE:
The Ethernet adapter has been set to the IP address 192.168.1.15
(If you prefer to use your primary Ethernet connection go to 3.3.2 below).

3.3.2 Setting an Alternative Configuration for the Primary Ethernet Port

Proceed with the steps shown in section 3.3.1 and replace step 6 with 6a and step 9 with 9a as shown below:



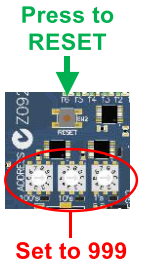
3.3.3 First-time direct Ethernet connection to FMEGATE

To connect to the configuration interface follow the steps below:

1. Ensure that the FMEGATE is powered up and operating (green LED 1 will be ON)
2. Connect the computer and FMEGATE Ethernet ports together with a cable.

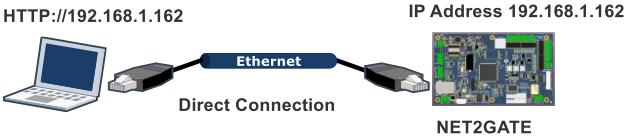
NOTE: Some older computers may require the use of a cross-over Ethernet cable for the connection. This can usually be determined if the LED indicators on the RJ45 socket are indicating that a link has not been established.

3. Set the FMEGATE address switches to “999”
4. Press the reset button on the FMEGATE



The reset process sets the Ethernet port to its default IP address 192.168.1.162

5. Open your web browser and enter the IP address of the configuration interface (192.168.1.162)

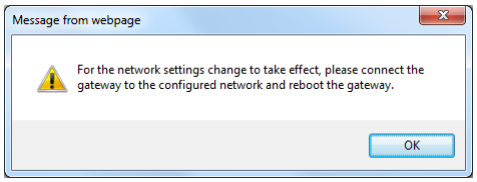


6. The Configuration Homepage will appear. As soon as any option is selected from the “homepage” the user will receive an alert requesting the address dials to be set to “000”. This change must be made before the user can proceed with any configuration options.

Use the configuration interface to change Date/Time Settings and Network Settings– see Sections 4.1 and 4.2

It is recommended that a User name / Password is now set to prevent unauthorised configuration changes– see Section 4.3

Continue from step 6 in section 3.2



7. Once configuration changes have been made the user will be advised that these changes will only take place after the FMEGATE has been reset.
8. Disconnect the Ethernet cable from the FMEGATE and press reset.

3.3.4 How to connect to a FMEGATE with an unknown IP address

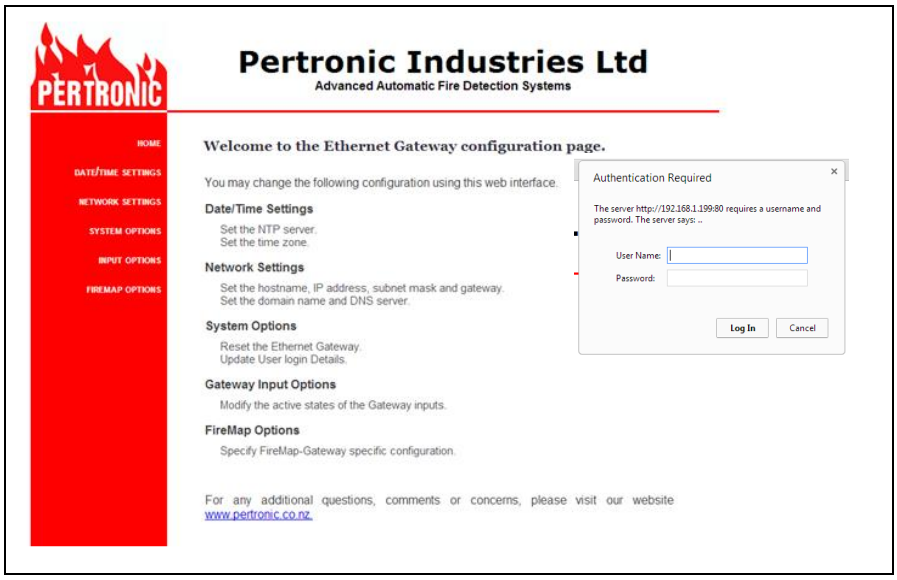
If a FMEGATE has been previously configured but its IP address is unknown it can be accessed to change its configuration by using the steps outlined in 0 .

4. CONFIGURATION INTERFACE

Once Network settings have been correctly configured access is then possible via any pc that can connect to the local network which the FMEGATE is connected to.

Enter the IP address 192.168.1.162 into your browser window.

Once connected the Configuration Interface Home Page will be displayed. If a username / password has been configured then selecting any of the configuration options will bring up a logon window–a user name and password must then be entered to proceed with configuration.



4.1 Date / Time Settings

Use this option to set details of the Time Server and Time Zone. This is compulsory for FMEGATE cards which are connected via the RS485.

NOTE:

If a network time service is not available then a NTP application can run on the FireMap server. See section 9.

4.2 Network Settings

Use this option to configure:

- **Connection Settings**

  Hostname, IP Address, Subnet Mask, Gateway
- **DNS Settings (optional)**

  Domain Name, Server 1, Server 2, Server 3

If you are unsure of the correct information please consult with your network's IT administrator.

4.3 System Options

Use this option to:

- Change the User / Password
- Reset the FMEGATE software (Allows a "remote" reset which can be useful when diagnosing network issues).

4.4 Input Options

Use this option to define the “active” state for each of the 16 Isolated Digital inputs.

Each input can have a contact closing or opening to trigger the “active” state.

4.5 FireMap Options

4.5.1 PC Timeout Settings

This option is used to set a delay before FMEGATE reports loss of communications with FireMap® to its associated Fire Panel.

The delay can be:

Set to Zero

This disables the timeout feature and FMEGATE will never report communication faults to the associated Fire Panel.

Set to a value no greater than 4 hours

Attempting to set values greater than 4 hours will result in a message “Timeout is out of range. Maximum timeout allowed is 4 hours only”. The timeout will then default to 4 hours.

5. OTHER FUNCTIONS

5.1 Isolated Digital Inputs

There are 16 Isolated inputs available to interface outputs from auxiliary equipment and conventional Fire Panels.

The inputs are supplied as two groups of eight. For each group there is a common +24Volt connection namely +Vin1 or +Vin2. This arrangement allows two independent 24 volt power supplies to be used; one for each group of 8 inputs.

Alternatively if all 16 inputs are from the same equipment then +Vin1 and +Vin2 can be connected to the same power supply. If the FMEGATE is housed in the Fire Panel cabinet then often the 24 Volt power supply can be from the Fire Panel.

The installer has the option of choosing whether the “active” state is triggered by a contact opening or closing (see Section 4.4).

Note: Selecting open for “activation” means that in the normal state current is being drawn by the input.

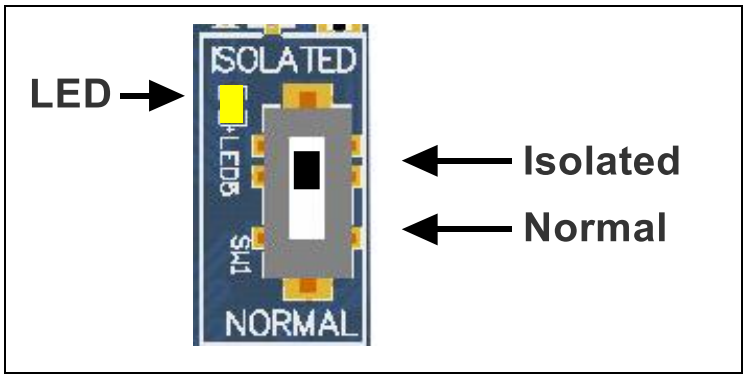
5.2 FMEGATE Isolation

There will be situations when service personnel need to “isolate” the FMEGATE output so that alarm conditions are not forwarded to FireMap® via the Ethernet port.

To isolate the output:

1. Switch the isolation switch to the ISOLATED position– the associated yellow LED will now be ON.
2. Return the isolation switch to the NORMAL position once service work has been completed– the yellow LED will now be OFF.

Note that an isolated FMEGATE card will show as an isolated device on any panel connected via the RS485 bus.



6. FIRE PANEL SETTINGS

The Fire Panel connected to the FMEGATE requires the following configuration changes:

1. Select the option “Network Card Attached”.
2. Increase the number of LCD Mimics connected by one.
3. LCD Mimic address 1 is used exclusively by the FMEGATE – no other LCD Mimics must use this address.

7. FIRMWARE UPDATE

The FMEGATE firmware can be loaded using a USB memory stick.

IMPORTANT

- The Firmware setup methods are different for v3.x.x and v4.x.x
- Due to architecture differences, boards containing v3.x.x firmware *cannot* be field upgraded to v4.x.x. This operation can only be performed in the factory. Once v4.x.x is installed, future v4.x.x firmware upgrades can then take place.
- The firmware update process does not check the installed firmware version, the uploaded file will always replace the currently loaded firmware.

7.1 V3.x.x Firmware Setup

USB memory stick setup:

- The USB memory stick must be FAT32 formatted, See Section 7.4 USB Flash Memory File System Format.
- Unzip the “InstallData” and “InstallScripts” folders from the FMEGATE upgrade zip file to the root of the USB stick file system.

FMEGATE setup:

- Check that the Isolate switch on the FMEGATE card is in the Normal position.
- Switch the USB select switch on the FMEGATE card to the Host position.
- Disconnect the Ethernet cable from the FMEGATE card Ethernet port (Note: This will generate a defect on any connected FireMap terminals).

Loading F220 keyboard/display firmware:

- Insert USB memory stick into socket J2.
- Wait a few seconds... The Isolate LED will blink during the update process, and remain on steady once the process has completed. (If a new version is not copied over either because the existing files are the same as or newer than those on the USB stick or the process did not start then the Isolated LED will blink once per second).
- Remove the USB memory stick
- Press the reset button SW2 “Reset”.
- Reconnect the Ethernet cable to the FMEGATE card Ethernet port.
- The FMEGATE will start using the new firmware.

7.2 V4.x.x Firmware Setup

USB memory stick setup:

- The USB memory stick must be FAT32 formatted, See S7.4 USB Flash Memory File System Format.
- Create a folder named “Net2Gate_Installer” in the root directory of the USB memory stick (if it does not exist already).
- Open the zip file containing the firmware to be installed. The zip file name will be in the format Net2Gate_v4.x.x.x.zip where x is the version numbering.
- From the Net2Gate_v4.x.x.x.zip file, copy the content of the folder called “Net2Gate_Installer” to the folder called “Net2Gate_Installer” on the USB memory stick.

FMEGate setup:

- Check that the Isolate switch on the FMEGATE card is in the Normal position.
- Disconnect the Ethernet cable from the FMEGATE card Ethernet port (Note: This will generate a defect on any connected FireMap terminals).
- Switch the USB select switch on the FMEGATE card to the Host position.
- Set DIP switch SW3 switch 04 to ON. This switch is only read at startup.
- Press the reset button SW2, “Reset”.
- Wait for the USB LED to start a slow blink to indicate the FMEGate is ready for a firmware upload. See Figure 7.2 USB LED Activity for F220 keyboard/display Firmware Upload.

Loading F220 keyboard/display firmware:

- Insert USB memory stick into socket J2.
- The USB active LED will be fast blinking while the firmware is being loaded. The firmware may take up to a couple of minutes to load and there may be brief periods of LED on time as loading progresses.
- The USB active LED will stay on continuously for at least five seconds after successfully loading the firmware from the USB memory stick. A slow blink at this stage indicates a failure.
- Remove the USB memory stick. The USB LED will go to a slow blink.
- Set DIP switch S3 switch 4 to OFF.
- Press the reset button SW2 “Reset”.
- Reconnect the Ethernet cable to the FMEGATE card Ethernet port.
- The FMEGate will start using the new firmware.

LED25 (USB Active)	Notes
Slow blink, 0.5s on /0.5s off |‾‾‾‾|____|‾‾‾‾|____	F220 keyboard/display is ready for a firmware upload and waiting for a USB memory stick to be inserted.
Constant fast blink |‾|_|‾|_|‾|_|‾|_|‾|_	F220 keyboard/display is uploading firmware from a USB memory stick. Brief interruptions to the blinking are normal.
Steady on ‾‾‾‾‾‾‾‾	Successful firmware upload.
Slow blink, 0.5s on /0.5s off |‾‾‾‾|____|‾‾‾‾|____	Firmware upload failure, if the USB memory stick is still inserted.

Figure 7.2 USB LED Activity for F220 keyboard/display Firmware Upload

7.3 Firmware Log

A log file will be created (if it does not exist already) on the USB memory stick and it will have brief details of the result of the last firmware upload timestamped and appended to the file.

To view the log:

- Plug the USB memory stick into a PC / Laptop and check that a log file has been written to the root directory. (The log file should contain the IP address of the FMEGATE as part of the file name–e.g. InstallLog192.168.0.21.txt).
- (Optional) Open the install log to view details of the installation.

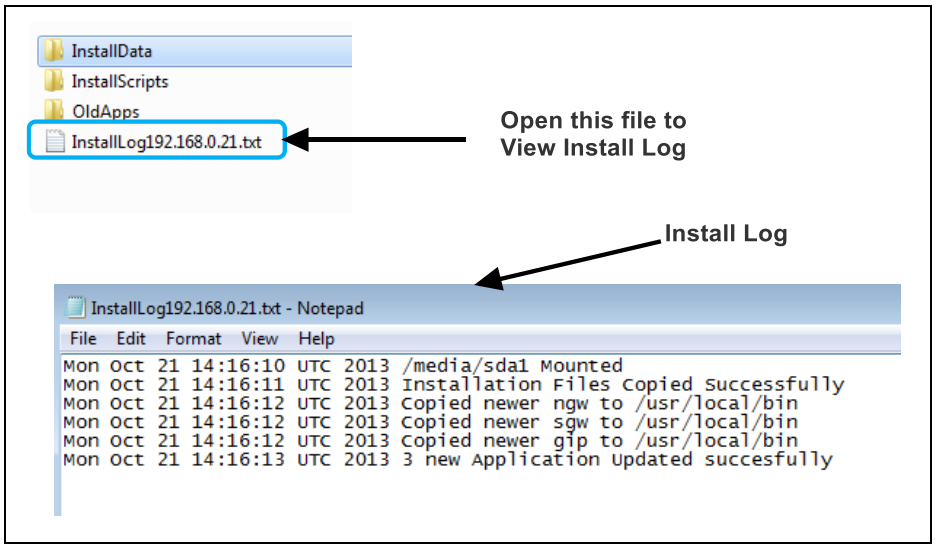


Figure 7.3 Example log

7.4 USB Flash Memory File System Format

The USB flash memory stick must have a FAT32 file system format for correct operation of the configuration process. The easiest way to check the format of USB memory is to plug the memory into a PC and when it is mounted:

- Open a file explorer window
- Right click the drive letter representing the USB memory
- Select “Properties” from the menu.

The file system should be FAT32.

If the file system is NTFS or some other format, it can be formatted to FAT32. There are several ways of formatting to FAT32, all involve total loss of any existing files. Any reformatting is best done on a smaller size (less than 32GB) USB memory stick. An internet search on “FAT32 USB format” should provide some good advice on reformatting using ordinary PC operating system utilities.

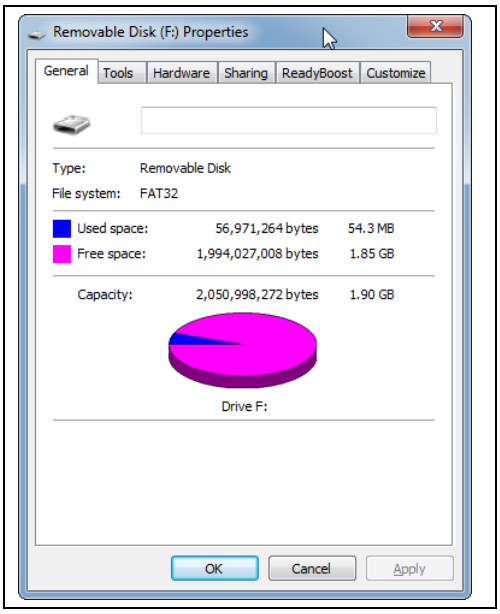


Figure 7.2 USB Flash Memory Stick File System Format

The configuration/firmware files must be loaded onto a USB flash memory stick for loading into the F220. The configuration files must have the correct names and be in the correct directory, the directory must also be in the correct location in the USB memory.

8. ORDERING INFORMATION

Product	Order Code
Ethernet Gateway MK2	FMEGATE
RS232 cable for FMEGATE	CL-NULLMODEM

9. TIME SERVER SETUP FOR ETHERNET GATEWAYS & PERTRONIC FIREMAP® SERVERS

Overview

Both FireMap and Pertronic Ethernet Gateways require access to an Internet time server.

If the FireMap system has been installed on its own dedicated network without access to the Internet then a separate Time Server application must be installed on the main FireMap PC (server).

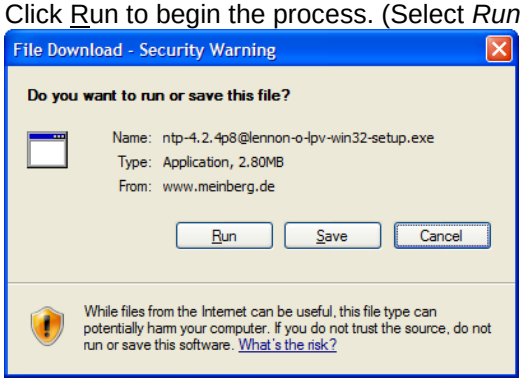
If, however, FireMap is sharing a network with other services then guidance should be sought from the network's IT manager relating to access to an Internet Time service.

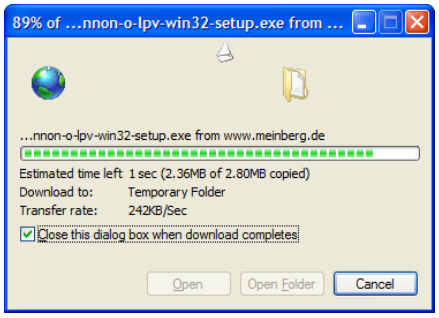
NTP (Network Time Protocol) *Time Server* application

NTP is an Open Source "time server" application. The latest version of *NTP* is available at *http://www.meinbergglobal.com/english/sw/ntp.htm*. Use the link, and follow the installation instructions. The steps are described in detail below:

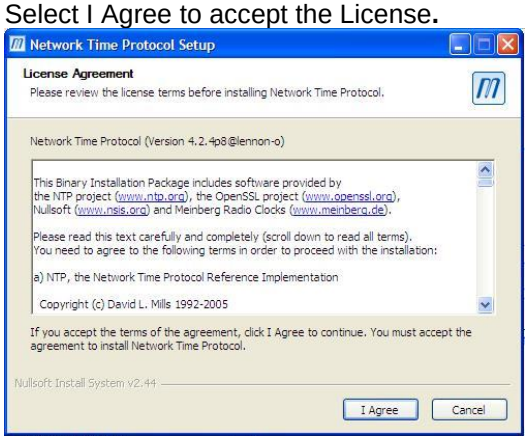
1. Select the link *ntp-4.2.6p5@lennon-o-lpv-win32-setup.exe.* **(as at 10/05/2013)**

2. Click Run to begin the process. (Select *Run* again, if the browser asks for confirmation.)

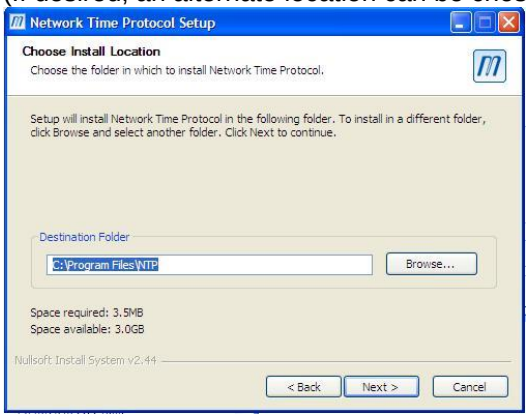




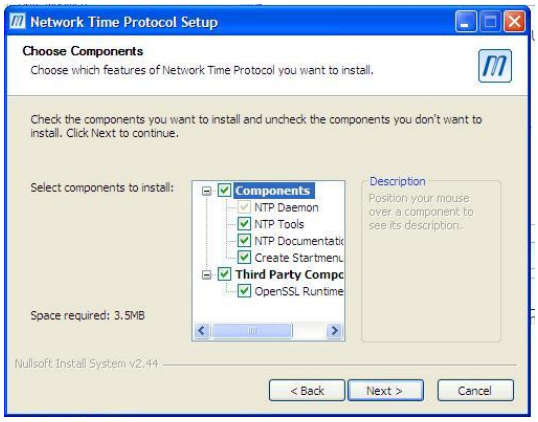
3. Select I Agree to accept the License.



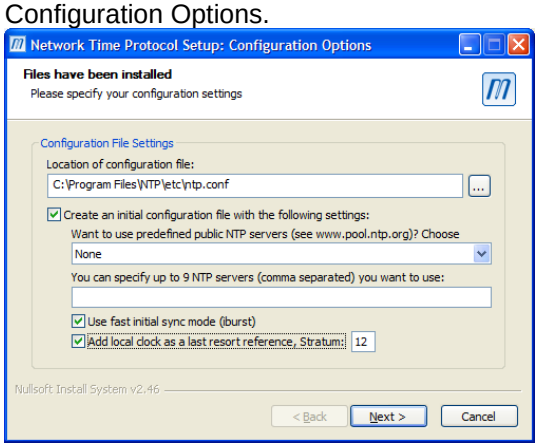
4. Click Next to accept the location in which to install the program.
(If desired, an alternate location can be chosen with the *Browse* button.)



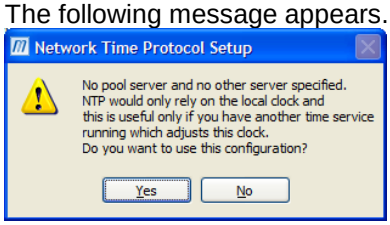
5. Click Next to accept the default Components that show in the dialogue box.
Files will be installed.



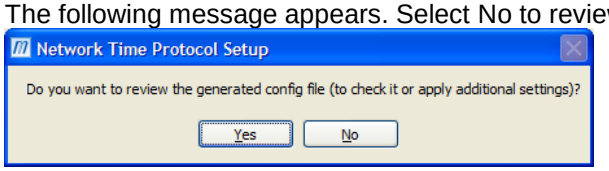
6. Make sure all boxes are ticked as shown below, and then click Next to accept the default Configuration Options.



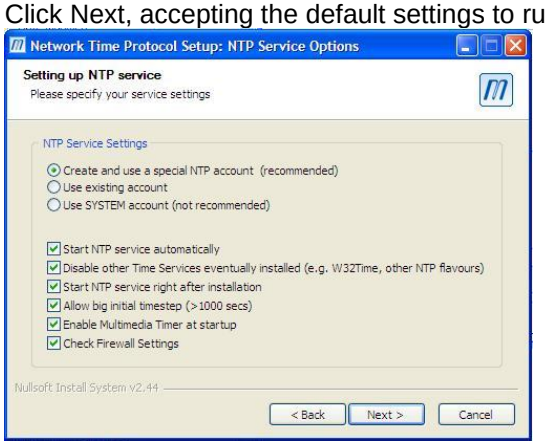
7. The following message appears. Select Yes to accept this configuration.



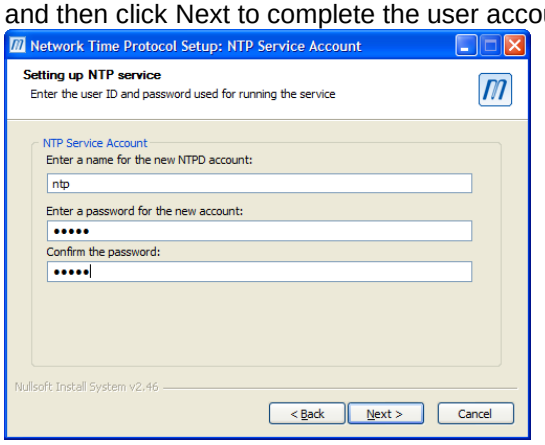
8. The following message appears. Select No to reviewing the config file.



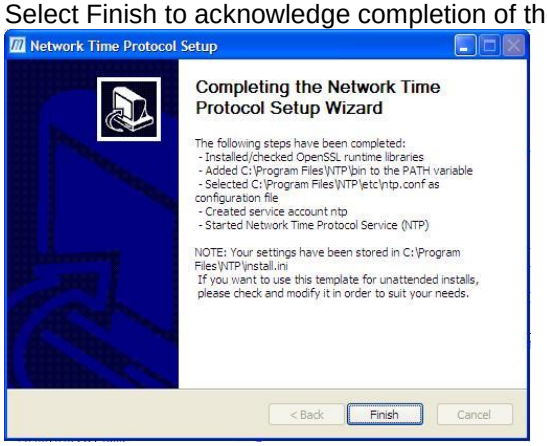
9. Click Next, accepting the default settings to run NTP in Windows.



10. Accept the default "ntp" user account name (or enter an alternative name). Create and confirm a password of 5 or more alphanumeric characters, and then click Next to complete the user account setup for NTP.



11. Select Finish to acknowledge completion of the NTP installation.



As previously noted, the network's IT administrator can provide advice on the use of NTP or any other time server application.

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New Zealand

Pertronic Industries Limited

Head Office
17 Eastern Hutt Road
Wingate
Lower Hutt
Phone +64 4 5673229
Fax +64 4 5673644
sales@pertronic.co.nz

Auckland
3 Gloucester Park Road
Onehunga
Auckland 1061
Phone +64 9 633 0226
Fax +64 9 633 0228

New Zealand website
www.pertronic.co.nz

Australia

Pertronic Industries Pty Limited

Melbourne
Unit B2
2A Westall Rd
Springvale
VIC 3171
Phone +61 3 9562 7577
sales.vic@pertronic.com.au

Sydney
Unit 9
38 South Stree
Rydalmere
NSW 2116
Phone +61 2 9638 7655
sales.nsw@pertronic.com.au

Brisbane
Unit 3
43-49 Sandgate Rd
Albion
QLD 4010
Phone +61 7 3255 2222
sales.qld@pertronic.com.au

Adelaide
65 Manton Street
Hindmarsh
SA 5007
Phone +61 8 8340 9533
sales.sa@pertronic.com.au

Perth
3/71 Beringarra Ave
Malaga
WA 6090
Phone +61 8 6555 3008
sales.wa@pertronic.com.au

Australian website
www.pertronic.com.au

USA

Pertronic Industries Inc

Phoenix
16597 N 92nd Street, Suite 110
Scottsdale
Arizona 85258
Phone +1 480 452 8330
Keith.Bowler@pertronic.com

Guilford
Connecticut 06437
Phone +1 203 598 1133
Raymond.Krickmier@pertronic.com

USA website
www.pertronic.com

12. DOCUMENT CHANGE HISTORY

Issue Number	Date	Reason for Change	Description of Change	Author
1	March 2014	First Issue		RDB
2 (NZ)	June 2014	CN 1764 DHCP added for Ethernet port for configuration purposes	Section 3.3 revised.	RDB
2.1 (NZ)	October 2014		Updated to use fixed default IP address. Setup using USB to Ethernet adapter.	RDB
2.5 (NZ)	29 April 2019	Correction	Corrected IP address error	JPF
3.0 (NZ)	2 May 2019	Firmware Programming	Added v4 Firmware programming.	SJR
3.0	5 March 2021	CN2954 Product name change. Updated RS232 cable product code.	Combined AUS & NZ manuals; updated product names; release of section 3.3.3 Ethernet address reset method.	JCB
3.1	12 Dec 2023	CN3399	Minor corrections and formatting changes	RJK